

uv Project Setup — Reference Sheet

Python venv / Jupyter kernel workflow using uv (Ubuntu, VS Code)

1. General Syntax (Template)

Use this pattern for any new project. Replace <python-version> and <kernel-name> with your actual values.

```
cd "/path/to/your/project"
uv init --python <python-version> --no-workspace --vcs none
uv add <package1> <package2> ipykernel
uv sync
source .venv/bin/activate
uv run python -m ipykernel install --user --name=<kernel-name> --display-name="Python (<kernel-na
```

--vcs none skips creating a .git repo. **--name** should be unique per project (no spaces) to avoid overwriting another project's kernel in Jupyter's registry.

2. Worked Example

Example: setting up a new 'practice' project under paper2, with pandas installed, Python 3.11, using a distinct kernel name to avoid clashing with the parent 'paper2' kernel.

```
cd "/home/thinkpad/Documents/BrototypeStudying/Paper two/paper2/practice"
uv init --python 3.11 --no-workspace --vcs none
uv add pandas ipykernel
uv sync
source .venv/bin/activate
uv run python -m ipykernel install --user --name=paper2-practice --display-name="Python (paper2/p
```

3. Using a Different Python Version (e.g. 3.9)

Check available versions first, install if missing, then follow the same template with a version-specific kernel name.

```
uv python list
uv python install 3.9

cd "/path/to/project"
uv init --python 3.9 --no-workspace --vcs none
uv add pandas ipykernel
uv sync
source .venv/bin/activate
uv run python -m ipykernel install --user --name=myproject39 --display-name="Python 3.9 (myprojec
```

4. After Setup — Select the Kernel in VS Code

1. Click the kernel name shown at the top-right of the notebook.
2. Choose 'Select Another Kernel' → 'Python Environments'.
3. Pick the entry matching your --display-name (e.g. "Python (paper2/practice)").

4. If it's not listed, click refresh, or run 'Developer: Reload Window' from the Command Palette (Ctrl+Shift+P).

5. Common Errors

ModuleNotFoundError after installing a package — usually means the notebook's kernel doesn't match the venv where you installed it. Reselect the correct kernel.

"Python Environment ... is no longer available" — the kernel is a stale reference to a deleted .venv. Rebuild the venv, then reselect (or re-register) the kernel.

bash: .venv/bin/activate: No such file or directory — the venv doesn't exist yet in this folder. Run `uv init` / `uv sync` first.

6. Cleanup

Remove all files created by `uv init` from a folder:

```
rm -rf .venv .git pyproject.toml .python-version README.md main.py .gitignore
```

List / remove named Jupyter kernels:

```
jupyter kernelspec list  
jupyter kernelspec uninstall <kernel-name>
```